MONASH University

FIT3143 Applied #1 Assessment (Week 6)

DISTRIBUTED WIRELESS SENSOR NETWORK

OBJECTIVES

- Analyse the design principles of distributed parallel computing systems (ULO3)
- Research and solve synchronisation problems typical in the design of parallel applications (ULO5)

MARKS

- The applied session is worth 10 of the unit's final mark.

APPLIED INSTRUCTIONS

1. Preparation is required for this Applied session. Please **DO NOT** plan to complete the session without any preparation/understanding.
2. Students should start working on the tasks at least 1 week prior to the assessed session.
3. Students must submit their answers and associated codes via Moodle in the required format before the start of the allocated class. A **late penalty** (5% per day) will apply if you do not submit the required materials on time!
4. In addition to regular offline/off-class marking, students' (or Team's) slides, answers, and/or source code will also be reviewed and assessed through in-class presentations, peer reviews, and/or question-and-answer sessions.
5. For **team/group presentations**, marks in general will be allocated for:
    - The correctness of the design, codes, or results
    - The quality and clarity of the presentation
    - The correctness of the explanation during the presentation

   The presentation content should include slides that are clear, easy to understand, and readable in a professional working environment (aka an industry setting). While the teaching team *<u>does not impose</u>* strict limits on the presentation format (slides/document format), nevertheless, if the presentation is hard to read/digest (e.g., speech is misaligned with the presented content, or slides/speech are unclear, or arguments are made without sound logic), marks will be deducted. Both the quality and correctness of your submitted/demonstrated work are assessed and graded.

6. For **individual/group interviews and Q&A sessions**, marks will be allocated for:
   - The correctness of the verbal answer
   - The quality of the verbal answer

   Focus on the most important idea in the answer. Marks will be awarded on the student's ability to explain their ideas with reference to the submitted code or documents in a concise, focused and accurate manner. Answers with errors, inaccurate or lengthy explanations will lose marks. Showing little to no understanding of the code will result in a loss of marks. Showing little to no understanding of the associated libraries, or submitting results will result in a loss of marks. Showing little to no understanding of how to select and filter information (aka dumping unfiltered answers without processing from the textbook/internet/AI agent) will lose marks.
7. Always make sure you read and follow the marking rubric well in advance of the deadline.
8. Marks will not be awarded if you skip the class, or do not make any submissions, or do not make any contributions in the working group/team, or make an empty submission to Moodle (unless a special consideration extension has been approved).
9. You are allowed to use Generative-AI to search for information and resources during the preparation period. However, you must declare in your report and upload all the prompt records (in PDF files).
10. AI tools are not allowed during the presentation period or any oral/coding interview sessions.
11. All submitted files should ideally include students' names, ID and Monash email addresses.

APPLIED IN-CLASS ASSESSMENT ACTIVITIES

Students are required to form teams of 2.

Note: For students who cannot form teams due to exceptional circumstances, arrangements must be made with the teaching team (generally the lead staff member in the corresponding applied/lab class) on/before Week 5.

❖ Team Preparation Period (before deadline): Divide the workload among team members and complete all the tasks. For each team, you must submit the required files (e.g., report, source code, slides, and corresponding documents) before the deadline.

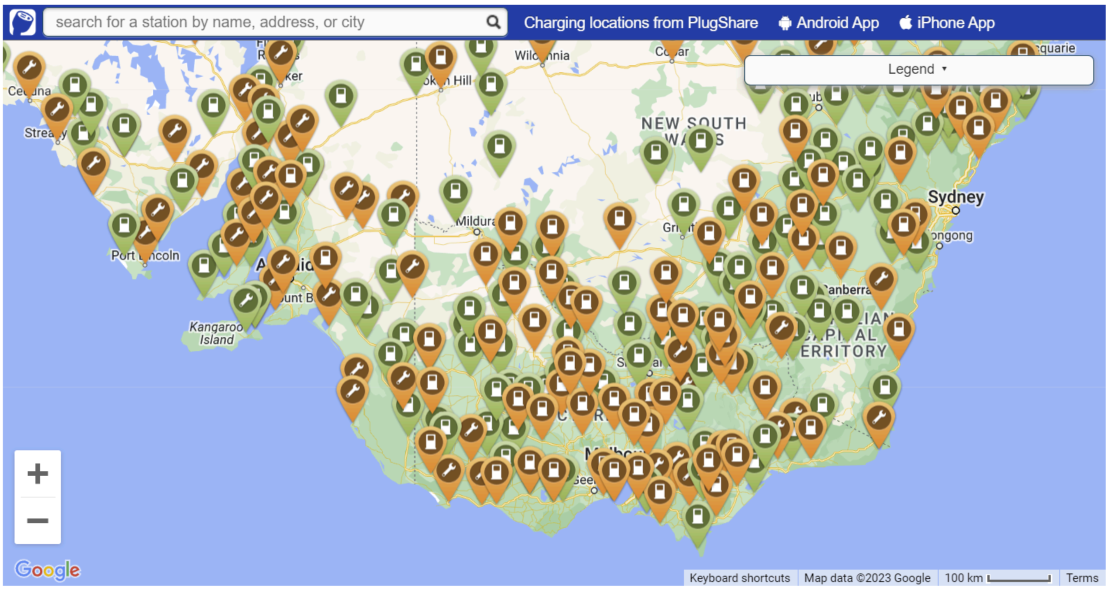
❖ Team Presentation Period (max 7 mins): For each team, you are required to present and demonstrate your design, code, results and observations during the session. All team members must be present in the applied session.

❖ Q&A Period (2 min): Each team member will be asked a few questions based on the submitted and/or presented work.

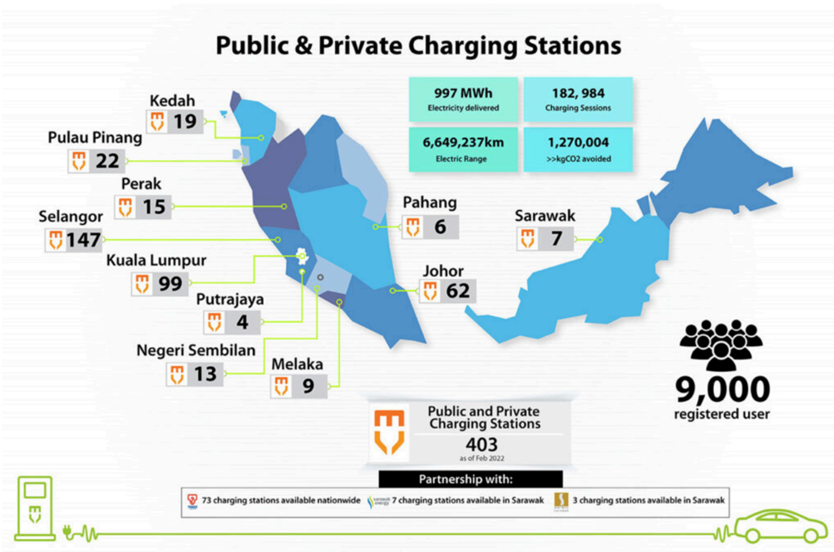
Note: If the required submissions are missing or any member(s) of the student team are absent, no marks will be given to the assessment.

Preamble

As the electric vehicle (EV) market continues to grow, the availability and accessibility of EV charging stations become critical factors in shaping the adoption and convenience of electric vehicles. EV charging station navigation refers to the ability to efficiently locate and navigate to nearby charging stations. Figures 1a and 1b illustrate the distribution of the EV charging stations in Victoria and in Malaysia. Effective navigation systems hold significant importance for optimal route planning and provide seamless urban mobility, thus contributing to the growth of the EV market and the transition to sustainable transportation.



(a)



(b)

Figure 1. Number of charging stations available in (a) Victoria and in (b) Malaysia. Images sourced from https://myelectriccar.com.au/ *https://www.mgtc.gov.my/what-we-do/low-carbon-mobility-2/chargev/*

To ensure the optimal selection of electric car charging ports, an Electric Vehicle Charging Navigation System (EVCNS) Centre will be coordinating the selection of the EV charging stations based on the real time information received such as the availability of charging stations and distance from the users.

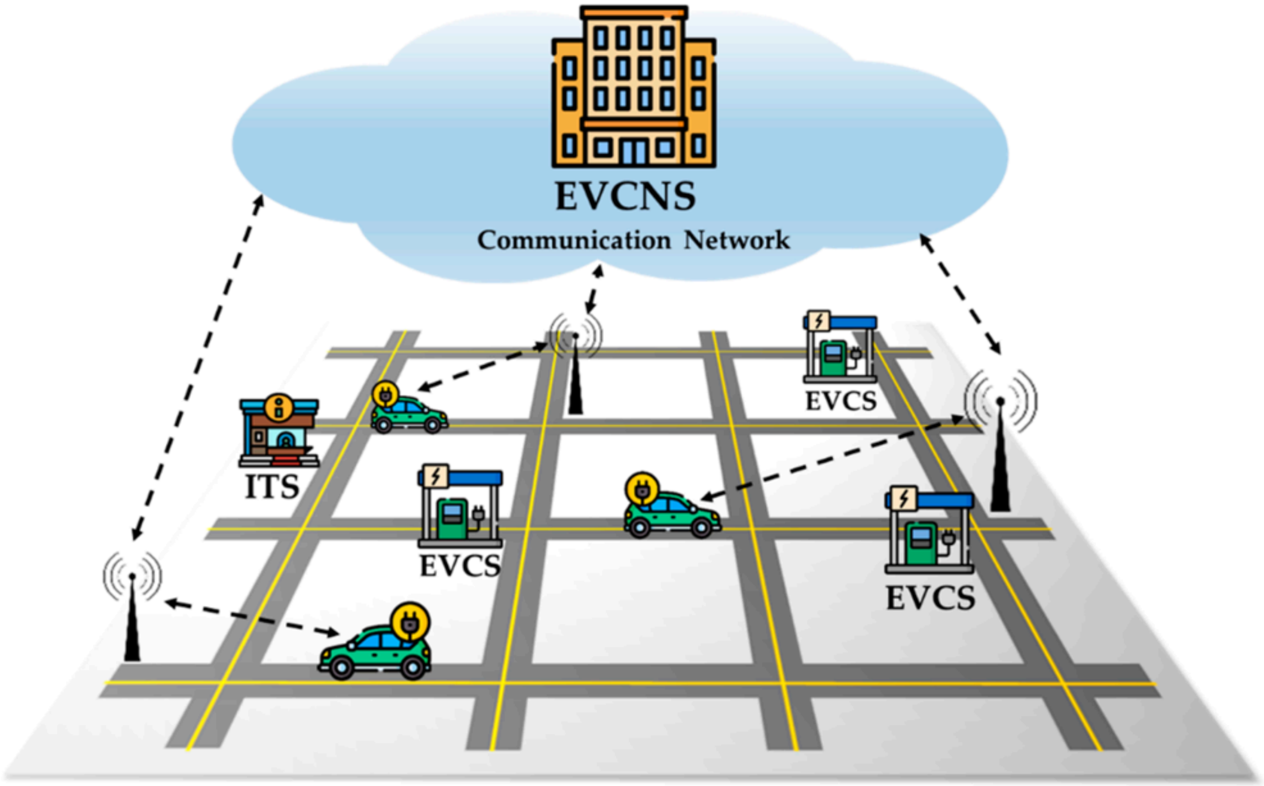


Figure 2. An electric vehicle charging navigation system (EVCNS) which consists of the EV charging station (EVCS) and intelligent transport system (ITS) that monitors the real-time traffic condition. Image sourced from https://www.mdpi.com/1996-1073/13/23/6255

Figure 2 illustrates an EV charging navigation system which consists of an intelligent transport system (ITS) and the EV charging stations. Data collected from EV charging stations could be transmitted to nearby EV charging navigation system centers or to a base station for further processing.

Objective of this assessment

In this assessment, we are attempting to simulate a wireless sensor network (WSN) of a set of interconnected EV charging nodes. Figure 3 illustrates an example overview of the WSN.

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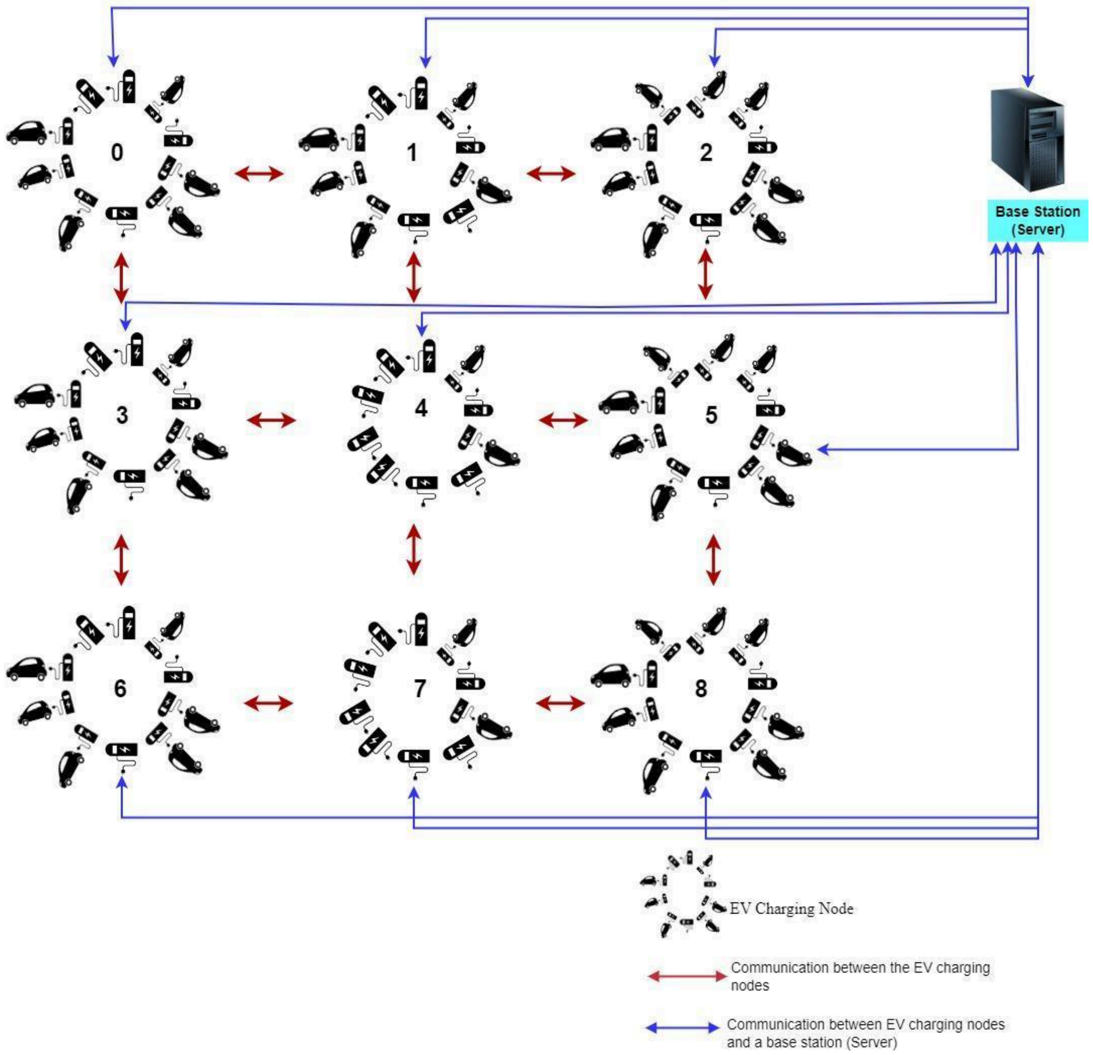

Figure 3. Overview of the wireless sensor network as the electric charging grid. In this illustration, the EV charging nodes are positioned in a 3 × 3 grid configuration. Each EV charging node has a number of in use or free charging ports. Each EV charging node communicates between the neighbours and the base station.

In this figure and for the purpose of illustration, nine EV charging nodes are positioned in a 3 × 3 Cartesian grid layout. The number of actual nodes will vary based on the area being observed. The charging nodes are positioned such that each node communicates with its immediate adjacent nodes.

Each EV charging node has a number of in-use or free charging ports. If majority of the ports (over the threshold) are in full use, the node will prompt for neighbour node data. If the received data from all the neighbours also show that their ports are heavily utilized (over the threshold), then the original node will alert the base station that the node and all its quadrant are almost being used up.

The base station receives the data from the nodes and logs the data. The base station will also be able to compute the closest available charging node (shortest distance from the original node) and alert the node to redirect all future incoming vehicles to the closest node. The aim of this individual assignment is to simulate the wireless sensor network (WSN) setup. A simulation software model needs to be designed using **Message Passing Interface (MPI)**.

Task 1 [Network topology]

In this task, you are required to compare and contrast different network topologies taught in Week 5. Based on your comparisons, choose the best type of topology to build the simulator for the wireless sensor network described in previous sections. Explain why.

Task 2 [Architecture design]

In this task, you are required to design the simulation software model architecture, with the assumption that software is going to be deployed to a cluster of multi-core machines. Your architecture must (at minimum) be using/based on **Message Passing Interface (MPI)**. In this task, you are required to design and draw your architecture using clear diagrams (e.g., UML Communication Diagrams) showing/indicating:

a) How your model would represent and simulate the base station,
b) How your model would represent and simulate all the charging nodes,
c) How you model would represent and simulate the charging ports (within each node),

Additionally, you are also required to clearly explain how the objects in your model would interact with each other and all the message passing needs. You should also explain how many machines (and associated cores) your program would need and the required network bandwidth.

Task 3 [Analysis]

In this task, you are required to perform communication analysis on your designed software architecture. Assume your software runs on a cluster connected via 1 Gbps links. For each type of message described in your architecture:

a) How would the transmission delay scale when the number of charging nodes increases?
b) Would increasing the number of base stations (servers) reduce the transmission delay?

Task 4 [Presentation slides and documentations]

In this task, you are required to prepare presentation slides (or documentation) to report and present your design, answers, and results from previous Tasks. You will have a maximum of 7 minutes to present your slides and approximately 1 to 2 minutes for the Q&A session.

Your presentation slides (or documentation) should have the following sections and content (at minimum):

a) (Task 1) Network topology

   This section should present your answers and solutions to Task 1.

b) (Task 2) Architecture design

   This section should introduce and present your simulation software model architecture. A clear diagram showing how you model the base station, the charging nodes, and the charging ports (within each node) is required to be included in your presentation. How your software would simulate the wireless sensor network (WSN) and what types of message passing are required should also be included in your presentation.

c) (Task 3) Analysis

This section should present the communication (complexity) analysis on your designed architecture. Additionally, you are also required to produce and present the following graphs (at minimum) for each type of messages used in your architecture design:

- Graphs showing how the transmission delay would change when the number of charging node increases
- Graphs showing how the transmission delay would change when the number of base station increases

d) Q&A [ETA: 1 to 2 min]
The section invites questions from the audience (TA/makers). The teaching team will ask questions related to your presentation and the submitted files.

Note: Practice makes perfect. If you are still deciding what to present and searching for information/content from the slides, chances are you will not be able to present within the time limit.
Note: You are allowed to put extra slides as an appendix (after the Q&A section) for your own convenience. Make sure they are clearly labelled so that markers will not be confused or mix them up with the other required sections.

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Submission Checklist

- Task 4 - Presentation slides (or documentation)
    - The documentation can be in the format of slides, docx, PDF or an equivalent format.
    - Please aim to keep the slides or documentation to within a 7-minute presentation (including the Q&A session).
    - Focus on emphasising the design, results and observations that would demonstrate a good understanding of the lab tasks.

- Task 1 to 3 – All associated detailed design documents, graphs, answers and results

- A separate AI declaration (in PDF format), **<u>if not</u>** already declared in the submitted documentation as aforementioned.

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INSTRUCTIONS FOR ALTERNATIVE ASSESSMENT

This section only applies if you or your team member(s) are **granted a special consideration extension** by the special consideration team (i.e., with a valid approval notice).

1. As a reminder, note that the regular 2 day extensions would not apply for applied/lab sessions (as they are in-class assessments with group/team works).

2. Instead of reviewing student's answers (in-class) in a presentation, peer review, and/or question and answer format during the regular in-class session, **student's answers (Task 4) will be assessed based on a recorded presentation video.** The video is required to be less than 9 minutes, and markers will be marking the assessment offline.

3. Markers will only mark the first 9 minutes of the video at regular playback speed. If students decide to compress the video by speeding up the playback speed, presentation quality could be affected and marks may be deducted according to the marking rubric.

4. Instead of having an in-person Q&A sessions, students will need to answer the following questions in their recording:
   i. Is your design (simulation software) closely resembling the architecture of a realistic wireless sensor network? Why/Why not? Can it be easily ported from a cluster environment to a distributed wireless sensor network in real life?
   ii. Does your implementation also consider using POSIX threads (or Open MP) to further enhance parallelism? Why/Why not?

5. Apart from the recorded video file (mp4), students are still required to submit the presentation slides (PDF/pptx), associated source codes (if any), and any AI declarations/reports (PDF) listed in the regular submission checklist via Moodle before **the approved extended due date**.

6. **A late penalty** (5% per day) will still apply if you do not submit the required materials on time (against the new extended date)! As a reminder, if you submit 7 days after the extended due date, you will still receive a 0 mark.

7. AI tools are generally not allowed during the recording sessions.

8. Before you start your presentation, you are required to show your face and your student ID in the video.